

Case Study/Assignment Topics from Computer Networks and Mobile Network

- ✓ You can select any one topic from the following list or any new topic related to our course.
- ✓ Study, understand and prepare the report for uploading in VTOP for assessment.

Important Topics/Areas in Computer Networks:

- **5G and 6G Networks**: Researching 5G/6G protocols, 3D beamforming, spectrum efficiency, and IoT connectivity optimization.
 - **Software-Defined Networking (SDN) and NFV**: Investigating SDN security, network function virtualization (NFV), traffic management, and network orchestration for 5G.
 - **Network Security and Privacy**: Exploring AI-powered intrusion detection, ransomware defense, zero-trust architectures, and security in edge computing.
 - **Internet of Things (IoT) and Industrial IoT**: Developing lightweight security protocols, device-to-device communication, and low-power network management.
 - **Edge and Cloud Computing**: Managing mobile edge caching, service migration, and multi-cloud technology efficiency.
 - **Artificial Intelligence for Networking**: Applying machine learning (ML) and deep learning to traffic forecasting, anomaly detection, and autonomous network management.
 - **Wireless and Wireless Systems**: Focusing on vehicular ad hoc networks (VANETs), Underwater Sensor Networks, and Wireless Body Area Networks.
 - **Energy-Efficient Networking**: Developing green network techniques and green communication algorithms.
 - **Blockchain for Networking**: Decentralized security and data integrity.
 - **Quantum Networking**: Secure communications.
 - **Traffic Engineering**: SDN-based routing optimization.
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- ☐ Secure and control sensitive data also in cloud environment (any)
 - ☐ The future of IoT and also bio metrics
 - ☐ Software defined networking

- ☐ Network security and also cryptography
- ☐ Network Function Virtualization
- ☐ Cognitive computing and also machine learning
- ☐ Micro services architecture
- ☐ Adaptive security
- ☐ Augmented and virtual reality
- ☐ Cloud networking
- ☐ Big data analytics in mobile networking
- ☐ Smart personal assistants
- ☐ Wearable's in sensor networks
- ☐ Blockchain as a service (BaaS)
- ☐ Containerization (traditional virtualization)
- ☐ Resource allocation SDN
- ☐ Ultra dense wireless networks planning
- ☐ SDN + Virtualized radio Access Networks also with Fog computing
- ☐ Spectrum efficiency enhancement by LTE-U also with Wi-Fi
- ☐ 5G wireless backhaul networks
- ☐ SDN based Elastic optical networks also in cloud.
- ☐ Green mobile cloud network: Green cloudlet
- ☐ C-RAN: Cloud Radio Access Network
- ☐ 5G networks multicasting
- ☐ Traffic engineering also in software defined networks
- ☐ D2D communication in 5G
- ☐ Over Wi-Fi secure device-to-device communication
- ☐ Cloud Robotics
- ☐ 5G networks for visible light communication
- ☐ Big data in mobile cloud networks
- ☐ Prevention and also in detection of network attacks
- ☐ SDN network automation to 802.11ac and also in IPv6

Recent Trends in Mobile Computing

- The following are the major research trends and areas in virtual reality and mobile web applications
- Mobile and wireless research integration
- Cloud computing and mobile crowdsourcing
- Networks defined by software and wireless signal applications
- Mobile system resource management
- Mobile computing infrastructure support
- Designing mobile systems and increasing energy efficiency
- Drone technology and smart vehicles
- Mobile web and social networking support
- Drone networking technology and robotics
- Modelling and analysis of mobile data
- Wireless system and signal applications
- The following topics of research are based on edge-centric learning and federated learning which resource-efficient is high.
- Deep neural network real-time applications, hardware accelerators, and requirements
- Federated learning (resource-efficient) and deep mobile architecture compression
- Approaches of distributed deep learning-based model training
- Advanced edge and cloud offloading in partitioned networks
- Deep learning model based commodity processors optimization such as NPU, DSP, and GPU
- Deep learning and recurrent architecture based understanding and analysis of audio data
- Inference time resource management supported by operating systems
- Binary networks and other low pressure in neural networks quantization
- Deep learning-based mobile interaction and speech models
- Virtual reality audio and mobile version based deep convolutional neural networks
- Analysing the activities and behavioral changes of the users based on Deep neural approaches